Week 3 Reading: Energy and Work

Work is the energy transferred by a force acting through a distance.
W = F d cos(theta). A force at right angles to the motion does no work.

Kinetic energy: K = (1/2) m v^2. The work-energy theorem says the net work done on an object equals its change in kinetic energy.

Practice Problems

1. A 2 kg cart moves at 3 m/s. What is its kinetic energy?
2. A 40 N force pushes a box 5 m along the floor. How much work is done?
3. A ball is lifted 2 m. Which forces do work on it, and how much?